



D5 · DELIVERABLE

Project Plan

Incremental WBS that keeps screening running, the parsing work package, estimates, PERT/CPM, Gantt, sprints and a risk register including reputational risk.

How the rebuild is delivered in seven increments without stopping screening, with a measurable acceptance threshold for parsing and a schedule computed from the activity network.

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COURSE
Software Engineering & Project
Management
B.Tech CSE · Batch 2024–28 · Semester
V

DOCUMENT
D5 · v1.0
1 October 2026 · ITM Skills University

1 Project charter

Project	Rebuild of the applicant-screening system as HireSense
Sponsor	Head of Talent
Problem	Unexplainable ranking that tracks university tier; résumé parsing fails silently on 29% of documents; no rejection log; no bias audit
Objectives	Explicit, configurable, versioned criteria · every automated rejection logged with a reason · parse failures never silent · recurring bias audit with a defined consequence · screening never stops during the rebuild
Measurable success criteria	Parse success $\geq 95\%$ and silent failures $\leq 1\%$ on the evaluation corpus · 100% of decisions logged with reason and criteria version · monthly audit on time 100% · zero days without screening · all Must requirements in the SRS verified
Scope	Parsing, criteria, screening, ranking, decision log, review queue, bias audit, legacy shadow comparison
Out of scope	Interview scheduling, offers, job board, payroll
Constraints	Case-study figures used as given; 90,000 applications/year; legacy must keep running until cut-over
Team	1 project lead / scrum master · 2 engineers · 1 QA · 0.5 talent-ops SME · 0.25 audit/compliance lead

2 Incremental delivery: screening never stops

The rebuild follows the **strangler-fig** pattern. Each increment is shippable on its own and adds a capability *around* the legacy system before replacing any part of it. Legacy stays the system of record until the increment's exit criteria are met, and every role has a rollback switch back to legacy.

INCREMENT	DELIVERS	SCREENING DURING THIS INCREMENT	EXIT CRITERIA
I0 Foundations	Evaluation corpus with ground truth; append-only decision log; legacy decisions shadow-logged	Legacy only (now logged)	Legacy reproduces 71% on the corpus; 100% of legacy decisions logged
I1 Stop silent rejects	Confidence gate wrapped around the <i>legacy</i> parser	Legacy + gate: low-confidence rejects go to review	0 automated rejections at confidence < 0.6
I2 Parser v2 + OCR	Layout-aware parser, anomaly cap, OCR fallback (WP-P)	Legacy primary; v2 in shadow	$\geq 95\%$ correct, $\leq 1\%$ silent on the corpus
I3 Criteria + routine	Versioned criteria, editor, preview; specified routine; rank keys; explain view	Canary: one role on HireSense, others legacy	Canary role: 2 weeks, review queue age < 2 days, no Sev-1 defects
I4 Bias audit	Audit service, specification (D4), scheduler, breach procedure	Canary + audits on all roles (shadow)	Two scheduled audits completed; breach drill passed
I5 Review + AI assist	Review queue, verified re-screen, grounded AI extraction	Canary	Median review time recorded; no ungrounded values accepted
I6 Parallel run + cut-over	All roles on HireSense with legacy in shadow; then decommission	HireSense primary, legacy shadow, then legacy off	Two audit cycles pass; sponsor sign-off

3 Work breakdown structure

Deliverable-oriented WBS. Level-1 elements follow the increments; work packages map to the schedule activities (§6). The parsing work package **WP-P** is defined in §4.

WBS	ELEMENT / WORK PACKAGE	INCREMENT	ACTIVITY	DELIVERABLE
1	Project management	all	—	Charter, plan, status reports
1.1	Charter and baseline metrics	I0	A	Signed charter
1.2	Sprint ceremonies, risk and change control	all	—	Sprint reviews, risk register
2	Foundations	I0		
2.1	Evaluation corpus and ground truth	I0	B	Corpus (legacy reproduces 71%)
2.2	Append-only decision log and shadow logging	I0	C	decision_log, triggers
3	Parsing (WP-P)	I1–I2		
3.1	Confidence gate around the legacy parser	I1	D	No silent rejects in production
3.2	Parser v2: layout, sections, dates, confidence	I2	E	parser v2.x
3.3	OCR fallback for scanned PDFs	I2	F	ocr.ts
3.4	CI evaluation gate	I2	E/F	Blocked releases below threshold
4	Criteria and screening	I3		
4.1	Criteria model, versions, editor and preview	I3	G	criteria_versions, editor
4.2	Specified routine, rank keys, explain view	I3	H	routine.ts, order.ts
4.3	Canary on one role	I3	I	Canary report
5	Bias audit	I4	J	Audit service, D4, breach drill
6	Review and UX	I5	K	Review queue, AI assist
7	Transition	I6		
7.1	Parallel run on all roles (2 audit cycles)	I6	L	Parallel-run report
7.2	Cut-over, legacy decommission, handover	I6	M	Sign-off

4 Work package WP-P: parsing quality

The case study gives the current parsing quality as a **29% failure rate** and asks for an acceptance threshold, plus the effort to reach it, defined as a work package.

FIELD	DEFINITION
Baseline	Legacy parser: 71% correct, 29% failing; every failure is silent (no confidence, no error)
Correct parse	All decision fields right: skills ($F1 \geq 0.8$), years of experience (± 1 year) and city, against ground truth
Acceptance threshold	$\geq 95\%$ correct on the evaluation corpus, and $\leq 1\%$ silent failures (wrong but confidence ≥ 0.6), and $\geq 90\%$ correct in every layout class
Measurement	seed/evaluate.ts on every build; release blocked in CI below threshold
Owner / duration	Engineer 1 · activities E + F · 29.0 working days expected (PERT)

TASK	EFFORT (PERSON-DAYS)	OUTPUT
Evaluation corpus with layout classes and ground truth	4	2,000 résumés; 29% hard layouts
Reading-order reconstruction (two-column detection)	6	pdf.ts readingOrder
Section headings: synonyms, letter-spacing	3	parser.ts headings
Date-range grammar (7 formats + OCR variants)	3	dates.ts
Confidence model, anomaly cap, gate	3	WEIGHTS, ANOMALY_CAP
OCR fallback (rasterise + PP-OCrv6) and engine benchmark	5	ocr.ts; PP-OCrv6 vs tesseract.js
Evaluation harness + CI gate	3	evaluate.ts

TASK	EFFORT (PERSON-DAYS)	OUTPUT
Regression tests for every defect found	2	parser.test.ts
Total	29	≈ activities E + F in the schedule

BASELINE CORRECT 71.0% legacy	ACHIEVED CORRECT 99.0% target ≥ 95%	SILENT FAILURES 0.30% target ≤ 1%	WORST LAYOUT CLASS 88% target ≥ 90%
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Status: 2 of 3 acceptance criteria met. Overall correctness ✓ · silent failures ✓ · every layout class ≥ 90% × (unusual dates: 88%).

The gap is season-style dates ("Summer 2023 – Spring 2025"), which the date grammar does not read. Of the 17 failures in this class, the anomaly cap routes 14 to review; 3 remain silent (a partially read date range).
Remaining task: add a season grammar (≈ 1 person-day), then re-run the evaluation gate.

Without OCR, v2 reached 91.1% overall. The 8% of scanned résumés made the OCR task necessary for the 95% target.

5 Effort and cost estimation (COCOMO)

Basic COCOMO, organic mode (a small team, familiar domain), sized from the **actual prototype**: 4,805 non-blank, non-comment lines in the API, seed generator and web client. Generated UI-library code and tests are excluded.

QUANTITY	FORMULA	VALUE
Size	KLOC	4.80
Effort	$E = 2.4 \times \text{KLOC}^{1.05}$	12.5 person-months
Duration	$D = 2.5 \times E^{0.38}$	6.5 months
Average staff	$E \div D$	1.9 people
Cost	$E \times ₹1,25,000 / \text{person-month (blended, assumption)}$	₹15.6 lakh

Cross-check against the bottom-up plan. The PERT schedule (§6) has an expected critical path of 83 working days (≈ 4.0 months). With about 4 FTE (1 lead, 2 engineers, 1 QA, plus part-time SMEs), that is ≈ 15.8 person-months, against COCOMO's 12.5 person-months over 6.5 months. The two estimates agree within 27%. The plan is shorter than COCOMO's duration because it uses more people in parallel than COCOMO's 1.9. COCOMO is calibrated on conventional projects, so it is used here as an independent sanity check.

6 Schedule: PERT and critical path

ID	ACTIVITY	INC.	PRED.	O	M	P	T _E	ES	EF	SLACK
A	Charter, stakeholder sign-off, baseline metrics	IO	—	4	6	10	6.3	0.0	6.3	critical
B	Evaluation corpus + ground truth (legacy reproduces 29%)	IO	A	6	10	16	10.3	6.3	16.7	critical
C	Append-only decision log + shadow logging of legacy	IO	A	6	8	12	8.3	6.3	14.7	4.3

ID	ACTIVITY	INC.	PRED.	O	M	P	T _E	ES	EF	SLACK
D	Confidence gate around legacy parser (stop silent rejects)	I1	B, C	4	6	10	6.3	16.7	23.0	18.7
E	Parser v2: layout, sections, dates, confidence (WP-P)	I2	B	12	18	28	18.7	16.7	35.3	critical
F	OCR fallback for scanned PDFs (WP-P)	I2	E	6	10	16	10.3	35.3	45.7	critical
G	Criteria model, versioning, editor + preview	I3	C	10	14	20	14.3	14.7	29.0	4.3
H	Specified routine, rank keys, explain-A-vs-B	I3	G	6	8	12	8.3	29.0	37.3	4.3
I	Canary: one role on HireSense, legacy in shadow	I3	D, E, H	10	14	20	14.3	37.3	51.7	4.3
J	Bias audit service, spec, scheduler, breach procedure	I4	H	8	12	18	12.3	37.3	49.7	6.3
K	Review queue + AI-assisted extraction	I5	F, H	6	10	16	10.3	45.7	56.0	critical
L	Parallel run, all roles (2 audit cycles)	I6	I, J, K	16	20	28	20.7	56.0	76.7	critical
M	Cut-over, legacy decommission, handover	I6	L	4	6	10	6.3	76.7	83.0	critical

$t_e = (o + 4m + p) / 6$, in working days. **Critical path: A → B → E → F → K → L → M.** Expected duration **83.0 days** ($\sigma \approx 4.6$), so there is a ~95% chance of finishing within 91 working days, by 2027-02-16. The project starts on 2026-10-12.

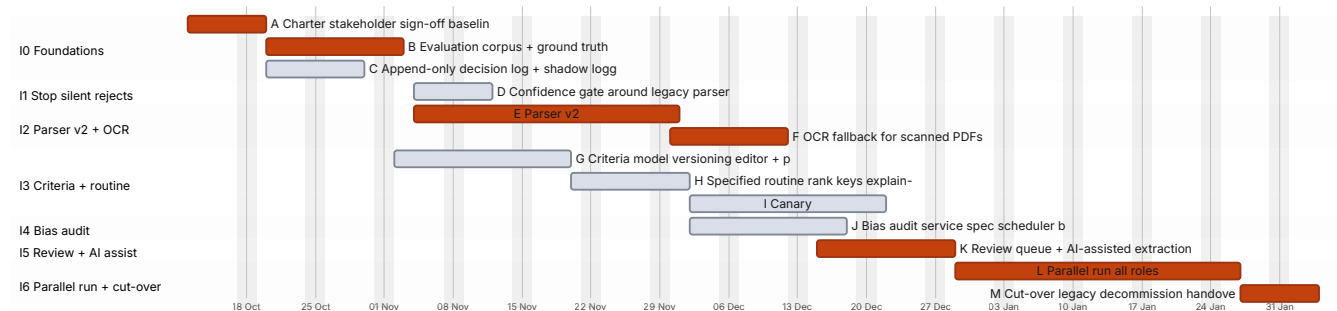


Figure 1. Gantt chart from the CPM early-start dates (working days; critical activities highlighted).

7 Agile execution: sprints, backlog, burndown

SPRINT	WEEKS	GOAL (INCREMENT)
S1	1–2	I0 foundations + I1 confidence gate
S2	3–4	I2 parser v2 (layout, sections, dates)
S3	5–6	I2 OCR + I3 criteria model and versions
S4	7–8	I3 routine, rank keys, explain; canary starts
S5	9–10	I4 bias audit + self-ID
S6	11–12	I5 review queue, AI assist, CSV export
S7	13–14	I6 parallel run all roles + CI gate
S8	15–16	I6 cut-over, decommission, handover

Product backlog

ID	USER STORY	PRIORITY	POINTS	SPRINT
US-01	As an audit lead, I want every automated rejection logged with a reason, so that any decision can be defended.	Must	5	S1
US-02	As a recruiter, I want unreadable résumés routed to a person instead of rejected, so that good candidates are not lost.	Must	5	S1
US-03	As engineering, I want an evaluation corpus with ground truth, so that parser quality is measured, not guessed.	Must	8	S1
US-04	As a recruiter, I want two-column and creative résumés parsed correctly.	Must	8	S2
US-05	As a recruiter, I want unusual date formats understood, so that experience is not undercounted.	Should	5	S2
US-06	As a reviewer, I want scanned PDFs read by OCR, so that the review queue shrinks.	Should	8	S3
US-07	As a criteria owner, I want to edit mandatory skills, bands, threshold, city and remote without a deploy.	Must	8	S3
US-08	As a criteria owner, I want every change versioned with a note, so that I can see which version decided what.	Must	3	S3
US-09	As a recruiter, I want to see why candidate A ranks above B.	Must	5	S4
US-10	As a criteria owner, I want to preview the impact and fairness of a change before publishing.	Should	5	S4
US-11	As an audit lead, I want a monthly bias audit with selection rates and impact ratios per declared group.	Must	8	S5
US-12	As an audit lead, I want a significant breach to pause automated rejection automatically.	Must	5	S5
US-13	As a candidate, I want to self-identify voluntarily, knowing screening never sees it.	Must	3	S5
US-14	As a reviewer, I want a queue sorted by age with the reason and the withheld outcome.	Must	3	S6
US-15	As a reviewer, I want AI-suggested field values that are grounded in the résumé text.	Could	5	S6
US-16	As an audit lead, I want to export the decision log as CSV.	Should	2	S6
US-17	As the Head of Talent, I want a parallel run on all roles before legacy is switched off.	Must	8	S7
US-18	As engineering, I want a CI gate that blocks a parser release below the acceptance threshold.	Must	3	S7
US-19	As the Head of Talent, I want a clean cut-over and handover.	Must	5	S8

Total **102 story points** over 8 sprints, about 13 points per sprint planned velocity. **Definition of Done:** code reviewed, unit tests and evaluation gate pass, decision log unaffected, docs updated, demo to the product owner.

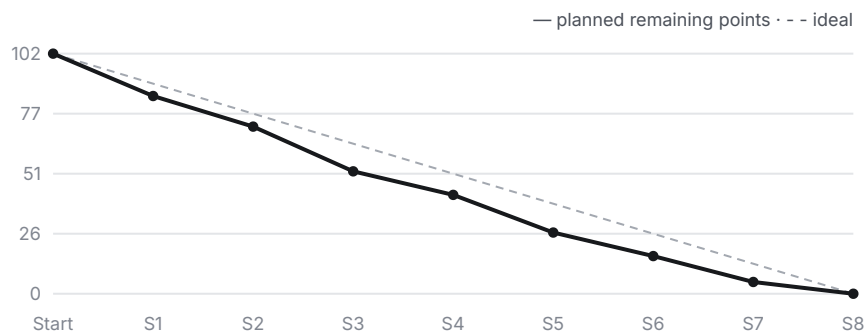


Figure 2. Release burndown (planned): remaining story points at the end of each sprint vs the ideal line. Actuals are recorded against this baseline as sprints complete.

8 Risk register

Probability (P) and impact (I) on a 1–5 scale; score = P × I. ≥ 15 high 8–14 medium < 8 low

ID	RISK	CATEGORY	P	I	SCORE	MITIGATION	CONTINGENCY	TRIGGER
R1	Reputational: a candidate or the press shows the screener discriminates (e.g. an age or tier proxy)	Reputational	3	5	15	Recurring audit; automatic pause on significant breach; documented job-relatedness; candidate-facing explanation	Pause auto-reject company-wide; public statement; external audit	Critical audit / complaint
R2	Silent parse failures continue during migration and reject qualified candidates	Quality	4	4	16	I1 ships the confidence gate in week 3, before any new parser	Route all rejects to review until gate is live	Silent-failure KPI > 1%
R3	Screening stops or slows during cut-over	Operational	2	5	10	Strangler increments; canary; legacy stays primary until exit criteria met; rollback switch per role	Flip role back to legacy	Queue age > 2 days
R4	Recruiters over-trust the new ranking (automation bias)	Human	4	3	12	Show reasons and paths, not just a rank; training; reviewers decide low-confidence cases	Sample audit of shortlists	Override rate ≈ 0
R5	Review queue overload after the gate goes live (29% of résumés)	Operational	4	3	12	OCR fallback (WP-P) cut the queue 215 → 94 in the prototype; staffing plan for reviewers	Temporary reviewers	Queue > 2 days
R6	Legal non-compliance with data-protection law for self-ID data	Legal	2	5	10	Separate table; audit-role access only; consent text; retention limits (DPDP Act 2023)	Suspend self-ID collection	Access-log anomaly
R7	Audit false alarms from small samples cause alert fatigue	Quality	3	3	9	n ≥ 30; rolling 90 days; significance test before pausing	Review thresholds quarterly	> 2 warnings/month with no cause
R8	Criteria change introduces new adverse impact	Quality	3	4	12	Mandatory preview + automatic audit on publish	Revert to previous version	Critical audit after change
R9	OCR / language model dependency unavailable or changes behaviour	Technical	2	3	6	Optional, with graceful fallback to manual review; pinned model versions; evaluation gate	Disable component	Eval drops below threshold
R10	Low self-ID response makes the audit blind	Quality	3	3	9	Explain purpose to candidates; track non-response as a KPI	Proxy-free analyses only	Non-response > 40%
R11	Decision log tampering or loss	Security	1	5	5	Append-only triggers; backups; access control	Restore from backup	Trigger violation logged
R12	Scope creep (interviews, offers)	Project	3	2	6	Scope fixed in charter; change control board	Defer to phase 2	Unplanned stories > 10%

9 Governance: RACI, communication, configuration management

ACTIVITY	PROJECT LEAD	ENGINEERS	QA	TALENT OPS	AUDIT LEAD	HEAD OF TALENT
Sprint planning & review	A/R	R	R	C	I	I
Criteria changes	I	C	I	R	C	A

ACTIVITY	PROJECT LEAD	ENGINEERS	QA	TALENT OPS	AUDIT LEAD	HEAD OF TALENT
Parser release (eval gate)	A	R	R	I	I	I
Monthly bias audit	I	C	I	C	R	A
Breach response	C	C	I	R	R	A
Cut-over decision	R	C	C	C	C	A

R responsible · A accountable · C consulted · I informed.

Configuration management and version control

- **Git** with trunk-based development: short-lived feature branches, pull requests with review, and CI running `bundle test` plus the parser evaluation gate.
- **Versioned artefacts:** the routine (code), the parser (`PARSER_VERSION` , stored on every parse), criteria (`criteria_versions` , stored on every decision), and the OCR and language models (pinned names). Any decision can therefore be reproduced exactly.
- **Change control:** criteria changes need a note and pass through preview and an automatic audit. Scope changes go to the project lead and sponsor.
- **Communication:** weekly status to the sponsor; a monthly audit report to the Head of Talent; incident notices within 1 working day of a critical audit.